

- 2 (a) $\frac{\text{kinetic energy of the child at Q with resistive force}}{\text{kinetic energy of the child at Q if there is no resistive force}}$

$$\begin{aligned} &= \frac{\frac{1}{2}mv_Q^2}{mgh} \\ &= \frac{\frac{1}{2}(4.8^2)}{9.81(7.0)} \quad \text{[M1]} \\ &= 0.168 \quad \text{[A1]} \end{aligned}$$

(b) percentage efficiency = $\frac{EPE}{KE} \times 100\%$

$$\begin{aligned} &= \frac{\frac{1}{2}kx^2}{\frac{1}{2}mv_Q^2} \times 100\% \\ &= \frac{54(2.1^2)}{20(4.8^2)} \times 100\% \quad \text{[M1]} \\ &= 51.7 \% \quad \text{[A1]} \end{aligned}$$

- (c) Any 2:

- Energy loss due to work done by resistive force / friction
- Sound produced due to collision with the soft board
- Deformation of the soft board